



Career expectations and outcomes: Evidence (on gender gaps) from the economics job market[☆]

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HIGHLIGHTS

- We investigate gender gaps in long-term career expectations and outcomes of PhD candidates in economics, by linking rich survey data to public data on job history and publication records.
- We find a robust gender gap in career expectations, with females about 10 percentage points less likely to ex-ante expect to get tenure or publish regularly.
- The gender gap in expectations is remarkably similar to the gap observed in academic outcomes.
- Expectations are similarly predictive of outcomes for males and females.
- Gender gaps in expectations can explain about 22 % and 14 % of the ex-post gaps in tenure and publications, respectively.

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ABSTRACT

This paper investigates gender gaps in long-term career expectations and outcomes of PhD candidates in economics. For this purpose, we match rich survey data on PhD candidates (from the 2008–2010 job market cohorts) to public data on job history and publication records through 2022. We document four novel empirical facts: (1) there is a robust gender gap in career expectations, with females about 10 percentage points less likely to ex-ante expect to get tenure or publish regularly; (2) the gender gap in expectations is remarkably similar to the gap observed for academic outcomes; (3) expectations are similarly predictive of outcomes for males and females, and (4) gender gaps in expectations can explain about 22 % and 14 % of the ex-post gaps in tenure and publications, respectively. In addition, leveraging variation in relationship status at the time expectations are reported, we show that: conditional on gender, (1) expectations regarding tenure and publications do not differ systematically by relationship status, and (2) the predictive power of expectations does not differ by the relationship status of the individual.

1. Introduction

While the gender pay gap in the US has declined substantially since the 1960s, the residual pay gap is remarkably persistent (Blau & Kahn, 2017). Gender gaps in high-skilled fields and occupations – where one might have expected convergence over time due to the increased representation of women in higher education – have proven to be surprisingly resilient. Examples include promotion gaps in the legal field (Azmat & Ferrer, 2017), wage gaps among business professionals (Bertrand, Goldin & Katz, 2010), and tenure gaps in academia (Antecol, Bedard & Stearns, 2018; Sarsons et al., 2021; Chen, Liu & Kim, 2022). There is also

a growing literature that shows expectations are an important predictor of educational choices and labor market outcomes (Giustinelli, 2023), and that these expectations often differ systematically by gender and can explain part of the underlying gender gap (Wiswall & Zafar, 2015; Reuben, Wiswall & Zafar, 2017; Cortés et al., 2023; Roussille, 2024).

In this paper, we investigate gender gaps in expectations and outcomes of PhD recipients in economics and adjacent fields early in their careers. For this purpose, we leverage novel survey data covering three cohorts from 2008 to 2010 (McFall et al., 2015). In these surveys, participants are asked about their career expectations over a number of

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outcomes, including receiving tenure and regularly publishing. We then link these participants to their eventual outcomes in terms of job history, publication records, and career citation counts as of 2022, resulting in a sample of 769 job market participants – 523 male and 246 female – over a 12–14 year horizon after receipt of the PhD.

We find that there are significant gender differences in career expectations early on. For example, 56 % of men think they are “very” or “extremely” likely to receive tenure, versus 45 % of women (p -value < 0.01). Likewise, 28 % of men versus 16 % of women in our sample believe that they will publish regularly in top journals (p -value < 0.01). These gender gaps remarkably mirror those that are observed in ex-post outcomes more than a decade later. For example, 56 % of men end up receiving tenure versus 44 % of women.

While average expectations and outcomes generally line up by gender, the natural question to ask is whether expectations are predictive of subsequent outcomes at the individual level. Here, we find that both tenure and publishing expectations are indeed related to better outcomes for both men and women (the relationship between publishing expectations and outcomes is generally weaker, especially for women, for whom the relationship is positive but not statistically precise). Given this, and the fact that expectations differ systematically by gender, we find that gender differences in expectations account for about 22 % of the gap in tenure achievement and 14 % of the gap in publication records. We find an even more pronounced reduction in the gender gap in post-PhD career citation counts, with expectations of tenure and publishing reducing the gender gap by 36 %. In research on drivers of gender gaps, conditional on standard observables, it is rare to find factors that can explain such a sizable part of the underlying gap (Buser, Niederle & Oosterbeek, 2014; Reuben, Wiswall & Zafar, 2017).

While our results are broadly consistent with earlier work in other settings that document gender gaps in expectations, and the empirical fact that they can explain part of the gaps in outcomes, it is somewhat surprising (at least to us) that we see similar results in this setting of high-ability high-ambition individuals. Our work is unique in the sense that there is little prior evidence linking expectations to outcomes over horizons beyond a few years. One paper that is closest to ours in methods and results is recent work by Azmat, Cuñat & Henry (2023) who find that aspirations of early-career lawyers differ substantially by gender, and can explain roughly half of the resulting gender promotion gap to partner. Our results, which leverage economics PhD recipients, both reaffirm previous findings in a new high-skilled occupation and provide novel evidence of expectation and outcome gaps in publications. While previous work has found gender gaps in publication and the credit received for those publications, to the best of our knowledge, there has not been any work that has examined gaps in publishing expectations and how they relate to outcomes.

One may expect career expectations and their relation to gender gaps to be intertwined with expectations regarding marriage, fertility, and household labor. Prior literature has shown that individuals make trade-offs between career and family, and that these trade-offs are significantly more costly for women (Goldin & Katz, 2008; Goldin, 2014; Juhn & McCue, 2017; Bursztyn, Fujiwara & Pallais, 2017; Fadlon, Lyngse & Nielsen, 2024). Additionally, Kuziemko et al. (2018) argue that women – particularly those with a college degree – significantly underestimate the impact of having a child on their labor supply, dubbed the “mommy effect”. To our knowledge, there has not been work that studies how relationship status influences ex-ante career expectations and their link to ex-post outcomes. Our unique setting and rich data allow us to investigate this directly.

Given that prior literature shows that choices of women in relationships tend to be constrained, one may expect single women to have better outcomes and to ex ante have higher expectations than women in relationships. On the other hand, women who are single in graduate school may end up in relationships later on. If they expect those future relationships to be constraining and if they fully internalize those constraints, then their expectations may not systemically differ from those

of women already in relationships. So, a priori, it is not clear what one may find.

Our analysis reveals that women in our sample who are in a committed relationship at the time expectations are elicited do not have significantly different expectations regarding tenure and publication achievement relative to their counterparts not in relationships.¹ Interestingly, we also find that there is no significant difference in tenure achievement for women in relationships compared to those who are single at the beginning of their career. We find similar results for men. Importantly, for both men and women, the predictive power of ex-ante expectations on their outcomes does not differ by relationship status.

Finally, the post-market survey also collects information on whether individuals in a relationship accepted the position that was their individually preferred option, or whether they made a different choice as a result of constraints imposed by their relationship. The sample size is small, but we find suggestive evidence that women were more likely to reject their individually preferred choice compared to men (15.4 % versus 11.6 %; p -value = 0.299). Additionally, men who accept their preferred position do not appear to have different expectations or outcomes compared to men who reject their preference due to a relationship, but women who reject their individually-preferred job are 14 percentage points less likely to believe they are likely to get tenure compared to women who did accept their preferred position. These gaps are reflected in outcomes, with men seemingly no less likely to obtain tenure by whether or not they accept their preferred position, while women who reject their individually preferred position are 16 percentage points less likely to obtain tenure compared to their peers who were not constrained by their relationship. This is broadly consistent with findings by Fadlon, Lyngse & Nielsen (2024) who find that initial labor market sorting has long term impacts for high-skilled women only. While our evidence is only suggestive given the small sample sizes, we believe this to be a critical area for future work.

2. Data

The structured nature of the economics job market, the near universal coverage of job market participants given public CVs and websites, and ten to twelve years of occupation and publication history present a unique opportunity to understand how expectations early in careers relate to long term, ex-post outcomes. In this section, we describe the data sources.

2.1. The job seekers' survey

The Job Seekers' Survey (JSS) was a longitudinal data-collection effort tracking the personal and professional trajectories of recent entrants to the junior PhD job market in Economics, spanning over three consecutive cohorts beginning with 2007–08. The survey was originally designed and administered at the University of Michigan by Brooke Helppie-McFall, Marta-Murray Close, and Bob Willis, and has been utilized in prior work concerning career satisfaction and joint location decisions (McFall et al., 2015; McFall & Murray-Close, 2016). The survey was administered to candidates from economics or adjacent PhD programs from the top 130 PhD granting institutions that were participating in the economics job market. In total, 1843 individuals were surveyed over the three cohorts. We believe that the JSS sample comprises nearly the universe of job candidates who expected to participate in the primary US job market for new economists in 2008–2010.

Each of the three cohorts was surveyed at multiple points across their career. All cohorts received a pre- and post-market survey. The

¹ To give some context, it is worth noting that individuals who are in relationships at the time of the survey seem to be positively selected in terms of baseline observable characteristics. In addition, a lower proportion of females are in relationships than men (73 % versus 79 %; p -value = 0.0567).

pre-market survey was administered in December of the job market year, by which time some interview invitations had been received but prior to the typical timeline for fly-outs or job offers. The pre-market survey aimed to capture individual characteristics, personal situations, and preferences over potential outcomes. The post-market survey was administered in August of the following year, after the job market had cleared, and participants had recently accepted and potentially started in their new position. The post-market survey collected information on outcomes and individual situation – including how relationships may have impacted their job search or outcomes.

The first and second cohorts also later received an early-career survey, in which career expectations were elicited. The early-career survey was conducted several years after the job market – 3 years after for the first cohort and 4 years after for the second cohort. For the third cohort, the original authors combined the post-market survey and the early-career survey. Thus, individuals in cohort 3 had their career expectations elicited right after graduation and at the beginning of their work careers, which is clearly preferable. We explore this mismatched timing in our robustness checks and find that it does not significantly alter our main findings.

Response rates average 64 % for the pre-market survey, 46 % for the post-market survey, and 39 % for the early-career survey. The response rate to the survey in which career expectations were elicited is 41 %. For clarity, we hereafter refer to the early-career surveys for the first and second cohorts and the post-market survey for the third cohort as the “expectation survey”.

2.2. Outcomes data

To understand whether career expectations are predictive of ex-post outcomes, we collected over ten years of data on both job and publication history for every participant. Ultimately, we were able to collect outcome data for 97 % of those for whom we had information regarding career expectations. We restrict all analyses to the sample with both expectations and outcomes data.

Job history was collected through both LinkedIn and publicly available CVs. Both sources were queried using APIs that took into consideration only public information – first and last names, PhD granting institution, and occasionally publications at the time of PhD attainment to aid in disambiguation. For both data sources, we often obtained multiple hits per survey participant. We then used programmatically assigned confidence scores to inform manual verification of the correct result. Publication data was accessed using Scopus, a publication database maintained by Elsevier. Scopus includes information on publication type, date of publication, journal of publication, coauthors, and citation count. A random sample of participants’ publication histories was spot checked against CVs to ensure accuracy.

Our job history data collection yielded information regarding employer, position title, duration, and location. This information was used to define positions as either academic or non-academic, and within academia as non-tenure track, tenure-track, or tenured. Tenured positions were identified by job title, location, and institution. Positions at institutions in the United States were determined to have tenure given titles including “Associate Professor”, “Professor”, “Full Professor”, etc. Positions outside of the United States often have different titles. For instance, the position of “Senior Lecturer” was determined to not hold tenure in the United States, but is a tenured position at Australian universities. Participants at institutions in the United States that utilized different position nomenclatures, particularly institutions where “Associate Professor” might not dictate tenure track, often included “with tenure” or “without tenure” in their titles.

2.3. Sample description

Our final analysis sample is restricted to individuals for whom we had information on both career expectations and ex-ante outcomes. These restrictions yield a final sample of 769 participants, 523 male and

246 female. The pre-market surveys occur from 2008 to 2010, following the Great Recession. How the recession impacted the relationship between expectations and outcomes is difficult to discern. Arguably, any impact the recession played on the expectations–outcomes link should not differ systematically across gender in our sample of highly-educated professionals, and should not pose a concern for our analysis on gender gaps. Full details pertaining to data cleaning can be found in Appendix B.

Regarding our sample, one concern might be that individuals who report expectations data are different from those who take the baseline survey but then do not report expectations data. This comparison is presented in the first two columns of Appendix Table A1. The analysis sample is more likely to be white (72 %) and a US citizen or permanent resident (47 %) compared to the overall surveyed sample, but is equally likely to be male (68 %) and from a top 30 university (57 %) – alleviating concerns of selection by gender or perceived ability. Column (3) of the table shows that the sample from whom we have both baseline survey data and expectations data is not very different from the analysis sample (which also includes respondents who take the expectations survey but not the baseline survey).

Across the different samples, we find that age at expectation survey, percent that accepted an academic position, job type preference, and program rank are strikingly similar, with statistically significant differences that are small in magnitude.² Appendix Table A2 presents additional information on selection, comparing survey recipient demographics across whether or not they completed the expectation survey. In this regard, we see that the subsample which completed the expectations module is more likely to be white, a US citizen, and be from the first cohort, compared to non-responders.

Table 1 presents means of key demographic, educational, and professional characteristics for the analysis sample. Whenever possible, we use participant characteristics from the expectation surveys, and leverage past surveys to fill in any missing data (see Appendix B for more information regarding data cleaning). Gender, relationship status, age, citizenship or permanent residency, and race are measured at the time expectations are elicited. Job preferences are elicited prior to the job market in the pre-market survey. The stated fields of the candidates were collected from the CVs of job market candidates at the time of the market. Finally, whether the initial job market placement is academic is asked in the post-market survey, but we complement this field when missing with our own data collection effort. Column (3) presents the average difference of these means by gender. Columns (6) and (9) present differences within gender by relationship status, measured as of the pre-market survey. Our results are robust to alternative measurement timings of relationship status.

Across gender, there are some key differences to note: men are 16 percentage points more likely than women to be white and are 8 percentage points more likely to list academic positions as their most preferred job market outcome. The difference in job preferences is driven by not-single men, as they are 14 percentage points more likely to prefer academia compared to their single counterparts, with single men preferring academia at comparable rates to women. Not-single men are also 17 percentage points more likely to be US citizens compared to their single counterparts and are more likely to have listed their primary field as an applied area. There are no statistically significant differences in any category by relationship status for women, though sample sizes are smaller.

² The significant difference in relationship status between the Baseline Survey Only sample and the other samples is largely driven by timing. Relationship status is measured in the Expectation Survey for column 1 which came 3–4 years after the job market for many of the individuals. For the Baseline Survey only sample, relationship status is measured at the beginning of the job market. Given that individuals are more likely to be in relationships as they age, this difference is not surprising.

Table 1
Descriptive statistics.

	Pooled			Men			Women		
	(1) Men	(2) Women	(3) Difference	(4) Single	(5) Not single	(6) Difference	(7) Single	(8) Not single	(9) Difference
Observations	523	246	–	110	413	–	67	179	–
Age at expectation survey	32.75	32.55	0.200	32.25	32.89	–0.638	32.31	32.64	–0.324
Accepted academic job	0.74	0.72	0.021	0.75	0.73	0.016	0.78	0.69	0.089
Holds US citizenship	0.49	0.43	0.054	0.35	0.52	–0.172***	0.37	0.46	–0.094
Race: white	0.77	0.61	0.161***	0.72	0.79	–0.071	0.57	0.63	–0.064
Job preference: academia	0.86	0.78	0.081**	0.75	0.89	–0.138***	0.75	0.79	–0.044
PhD from top program	0.56	0.59	–0.021	0.53	0.57	–0.047	0.55	0.60	–0.046
Field of study									
Applied	0.79	0.88	–0.084***	0.66	0.83	–0.164***	0.86	0.88	–0.025
Macro/trade/finance	0.45	0.34	0.111***	0.49	0.44	0.051	0.36	0.33	0.030
Econometrics	0.34	0.37	–0.030	0.32	0.34	–0.028	0.34	0.38	–0.034
Theory	0.19	0.11	0.081***	0.19	0.18	0.010	0.06	0.12	–0.059
Survey cohort									
Cohort 1	0.35	0.37	–0.016	0.38	0.35	0.036	0.30	0.40	–0.098
Cohort 2	0.26	0.27	–0.008	0.14	0.29	–0.157***	0.27	0.27	0.001
Cohort 3	0.39	0.36	0.024	0.48	0.36	0.121**	0.43	0.34	0.098

Sample is restricted to participants with expectation and outcome data for both tenure achievement and publishing. Job preferences and field of study are measured in the pre-market survey. Accepted position is measured in the post-market survey (which is the expectation survey for the third Cohort) and supplemented by our own data collection when missing. Age, relationship status, citizenship, and race are measured at the time of the expectation survey. The indicator for PhD from a top program is comprised of 39 U.S. and international programs, detailed in Appendix B. Single includes “Single” and “Dating”. Not Single includes “In a Committed Relationship”, “In a Marriage-Like Relationship”, and “Married”. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3. Empirical results

3.1. Expectations and outcomes

Career expectations are elicited via a six-point Likert scale, spanning “Extremely Unlikely”, “Very Unlikely”, “Somewhat Unlikely”, “Somewhat Likely”, “Very Likely”, and “Extremely Likely”. While expectations were elicited for several career outcomes, we focus our analysis on tenure achievement and publishing record. All career outcomes are asked in the same survey question, and begin with “How likely do you think it is that, **over the course of your life...**”. For tenure, we take the maximum of participant’s responses to “You will hold a tenured position at a research university?” and “You will hold a tenured position at a four-year college?”. For publishing, participants are asked their expectation of “You will publish regularly in top journals?”. We employ indicators for whether the respondent thinks the outcome is likely, where the indicator is equal to one if the response is either “very” or “extremely” likely.³ These kinds of questions suffer from the familiar issue that they may not be interpersonally comparable. For example, two people answering “very likely” may have very different subjective likelihoods in mind (Manski, 2004). However, we are constrained to work with the data that we have.

The methodology for determining whether a participant successfully obtained tenure is described above. The expectation question relating to publications was worded in a more ambiguous way, letting the respondent to independently determine what is meant by a “top” journal and what “regularly publish” means in terms of frequency. As a result, determining what is a successful career outcome for regularly publishing in top journals is more subjective. Our preferred measure which is used throughout our analysis is whether or not the participants published in a top general interest journal at any point after they received their PhD. The journals included in this measure are defined in Appendix Table B3.⁴ We also present evidence on the additional outcome of career citations in Appendix A. Career citations are defined as the sum of citations on academic publications, categorized by Scopus as either an

article or review article, on publications occurring after the completion of the participant’s PhD.

Given the richness of our novel dataset, we are able to analyze not only if there are gaps by gender in expectations and outcomes, but also whether there are any significant differences within gender by relationship status. Table 2 presents the means for the career expectation questions (Panel A) and ex-post outcomes (Panel B). We also present differences in these means between men and women, as well as differences by relationship status within gender. We first report the raw differences, and then report the differences including controls for demographics, field, and program quality.⁵

Looking at the first four columns in Panel A, we see there is a (statistically and economically) significant gender gap in expectations. On average, men are 10.9 percentage points more likely to believe they are likely to receive tenure, and 11.8 percentage points more likely to believe they will publish in top journals at baseline. Controlling for differences in baseline characteristics, we see the gaps are largely unchanged: they are 9.3 and 11.2 percentage points, respectively.

Turning to outcomes in Panel B, two things stand out: (1) the outcome levels are quite similar to the expectations (though, this is partly an artefact of how the variables were constructed), and (2) the gender gaps in expectations are closely mirrored in ex-post outcomes, with men 11.9 percentage points more likely to receive tenure and 10.4 percentage points more likely to publish in top journals.

The remaining columns of Table 2 cut the data further by gender and relationship status. Expectations for women do not differ statistically by relationship status. This is in contrast to other settings which document systematic differences in expectations for women by relationship status (Bursztyn, Fujiwara & Pallais, 2017). This trend also holds for men, with comparable expectations across relationship status. Turning to outcomes, they generally look more favorable for not-single individuals within gender. However, the differences are not statistically significant.

³ Note that these questions are fielded to all respondents, regardless of whether they had a tenure-track/academic job at the time of the survey or not.

⁴ Our results are robust to alternative measures of both publishing quality and timing; see Appendix Table A3.

⁵ The raw differences are obtained from a simple linear regression, where we report the coefficient on the relevant indicator – either if the participant self-identifies as female or if the participant reports being in a relationship in the survey in which expectations were elicited. The differences with controls come from corresponding regressions where we also control for the various characteristics. We account for missing data in characteristics by assigning missing values an arbitrary constant and including an indicator for if the value was missing.

Table 2
Expectations and outcomes by relationship status.

	Pooled				Men				Women			
	(1) Men	(2) Women	(3) Difference	(4) Difference with controls	(5) Single	(6) Not single	(7) Raw difference	(8) Difference with controls	(9) Single	(10) Not single	(11) Raw difference	(12) Difference with controls
Panel A: Expectations												
Tenure	0.56	0.45	−0.109*** (0.038) [−0.172, −0.046]	−0.093*** (0.034) [−0.149, −0.036]	0.55	0.56	0.007 (0.053) [−0.081, 0.095]	−0.031 (0.046) [−0.107, 0.045]	0.46	0.45	−0.016 (0.072) [−0.134, 0.103]	0.036 (0.066) [−0.073, 0.145]
Publish in top journals	0.28	0.16	−0.118*** (0.031) [−0.169, −0.068]	−0.112*** (0.031) [−0.163, −0.061]	0.29	0.28	−0.012 (0.049) [−0.093, 0.068]	−0.043 (0.050) [−0.125, 0.039]	0.18	0.16	−0.023 (0.054) [−0.112, 0.067]	−0.010 (0.058) [−0.106, 0.086]
Panel B: outcomes												
Tenure	0.56	0.44	−0.119*** (0.038) [−0.183, −0.056]	−0.118*** (0.036) [−0.178, −0.058]	0.49	0.58	0.085 (0.054) [−0.003, 0.174]	0.065 (0.053) [−0.021, 0.152]	0.43	0.44	0.009 (0.071) [−0.109, 0.126]	0.040 (0.071) [−0.077, 0.158]
Publish in top journals	0.32	0.22	−0.104*** (0.033) [−0.159, −0.049]	−0.104*** (0.033) [−0.157, −0.050]	0.28	0.33	0.052 (0.049) [−0.028, 0.133]	0.017 (0.048) [−0.062, 0.096]	0.15	0.25	0.097* (0.054) [0.007, 0.186]	0.098* (0.055) [0.008, 0.189]
Observations	523	246	769	769	110	413	523	523	67	179	246	246

Sample is restricted to participants with expectation and outcome data for both tenure achievement and publishing. Controls for difference specifications in columns (4), (8), and (12) include age, citizenship, race, fixed effects for cohort, if their most preferred outcome is academia, if their accepted job was in academia, PhD rank, and field of study. Job preferences and field of study are measured in the pre-market survey. Accepted position is measured in the post-market survey (which is the expectation survey for the third Cohort) and supplemented by our own data collection when missing. Age, relationship status, citizenship, and race are measured at the time of the expectation survey. The indicator for PhD from a top program is comprised of 39 U.S. and international programs, detailed in Appendix B. Top journals are defined as any of the following: The American Economic Review, Econometrica, Journal of Political Economy, Quarterly Journal of Economics, Review of Economic Studies, American Economic Journal: Applied Economics, American Economic Journal: Economic Policy, American Economic Journal: Microeconomics, Review of Economics and Statistics, Economic Journal, and Journal of the European Economic Association. Standard errors in parentheses. 90 % confidence interval in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3
Relationship between expectations and outcomes.

	Pooled			Men			Women		
	(1) Men	(2) Women	(3) Difference	(4) Single	(5) Not single	(6) Difference	(7) Single	(8) Not single	(9) Difference
Receive tenure	0.237*** (0.050)	0.326*** (0.075)	0.088 (0.089)	0.313** (0.122)	0.231*** (0.055)	−0.082 (0.129)	0.328** (0.152)	0.311*** (0.090)	−0.016 (0.169)
Publish in top journals	0.134*** (0.045)	0.064 (0.069)	−0.069 (0.082)	0.138 (0.101)	0.133** (0.052)	−0.005 (0.109)	0.012 (0.116)	0.070 (0.087)	0.058 (0.140)
Observations	523	246	769	110	413	523	67	179	246

The outcome variable is an indicator for if the participant achieved the relevant career outcome, as described in Section 3.1 and Appendix B.3. Each cell represents the coefficient from a separate regression on the indicator for if the participant indicated the outcome was either Very Likely or Extremely Likely. Columns (3), (6), and (9) are the respective differences between the coefficients. Sample is restricted to participants with expectation and outcome data for both tenure achievement and publishing. All specifications include controls for age, citizenship, race, fixed effects for cohort, if their most preferred outcome is academia, if their accepted job was in academia, PhD rank, and field of study. Job preferences and field of study are measured in the pre-market survey. Accepted position is measured in the post-market survey (which is the expectation survey for the third Cohort) and supplemented by our own data collection when missing. Age, relationship status, citizenship, and race are measured at the time of the expectation survey. The indicator for PhD from a top program is comprised of 39 U.S. and international programs, detailed in Appendix B. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

One exception is the likelihood of publishing in top journals for women, where non-single women are about 10 percentage points more likely to end up doing so.

3.2. Do expectations predict outcomes?

While the gender gap in both expectations and outcomes in our sample is interesting on its own, we are interested in whether ex-ante expectations are predictive of ex-post realizations at the individual level. For this purpose, we employ a simple linear probability model regressing the indicator for whether the relevant outcome was achieved onto an indicator that equals one if the participant ex-ante thought the outcome was “very” or “extremely” likely to happen, controlling for the vector of characteristics. Table 3 reports these estimates.

The predictive power in expectations is economically and statistically significant and comparable across genders. Men who believe that they are very or extremely likely to receive tenure are 23.7 percentage points more likely to actually receive tenure compared to their counterparts with lower expectations. For women, we find a similar association: women with higher expectations are in fact 32.6 percentage points more likely to receive tenure. While the difference between genders is not statistically significant due to power issues, the gap is economically significant at 9 percentage points.

For publishing, expectations remain highly significant predictors for men, leading to a 13.4 percentage point increase in the likelihood of publishing in top journals. For women, the relationship between expectations and outcomes for publishing is weaker (and statistically imprecise).

Instead of collapsing the Likert-scale expectations data to a binary variable, Fig. 1 shows that underlying variation in the data. Realizations generally are increasing for both men and women in their expectations, though the relationship is not necessarily monotonic. For example, 82 % of men and 71 % of women who report that tenure is “extremely likely” end up receiving tenure. On the other hand, the corresponding proportions for men (women) who report “very unlikely” for this outcome are 12 % (15 %).

Looking across the remaining columns of Table 3, we see that relationship status does not meaningfully affect the predictive power of expectations (within gender).⁶ A priori, given the evidence that women do not fully anticipate the employment effects of motherhood (Kuziemko et al., 2018), and that family–career trade-offs tend to be more binding

for women (Fadlon, Lyngse & Nielsen, 2024), one may have expected the relationship between expectations and outcomes for women to differ systematically by relationship status. On the other hand, if single women expect their future relationships to be constraining and already internalize those constraints, we may not find a systematic difference by relationship status. Our results seem to be consistent with the latter explanation.⁷

We can also investigate the link between the number of citations individuals receive and their ex-ante expectations, presented in Appendix Table A4. We observe relationships similar to those in Table 3, with a strong association between expectations and career citations in the pooled sample. While the coefficients on both expectations are large in magnitude, expecting to likely publish in top journals is associated with an increase of 110 career citations compared to an increase of 30 from expecting to receive tenure. Looking at differences by gender, the associations are (economically and statistically) larger for men.

3.3. Do expectation differences explain gender gaps in outcomes?

Given that there are gender gaps in career expectations and that these expectations systematically correlate with outcomes, a natural question to ask is the extent to which gender gaps in career expectations can account for career outcome gaps. To this effect, we estimate the following simple linear regression model:

$$\mathbb{1}[Y_{i,j}] = \beta \mathbb{1}[X_{i,j}] + \eta \mathbb{1}[female_i] + \gamma' Z_i + \theta_i + \epsilon_i, \quad (1)$$

where Z is a vector of individual-level controls including expectations. In Table 4, we present η , the coefficient on the indicator for the participant identifying as a female. Across the columns, we include a growing set of controls.

Column (1) presents the raw gender gaps in outcomes observed in our data – 11.9 percentage points and 10.4 percentage points for tenure and publishing, respectively. The inclusion of relationship status or standard controls (columns 2 and 3) has little impact on the gender estimate. The last two columns of the table show that expectations account for a non-trivial portion of the observed gender gaps in both tenure achievement and publication. For tenure, comparing column (5) with column (3), we see that expectations account for 2.5 percentage points, or about 22 % of

⁶ One exception is that the predictive power of expectations for actual tenure is weaker for not-single men (relative to single men), but the difference is not statistically significant at conventional levels.

⁷ There is another possibility: if females do not anticipate the effects of motherhood (of which there is evidence – Kuziemko et al. (2018)), then since most females – regardless of relationship status – have not yet had children in graduate school, we would not see differences in the predictive power of expectations by relationship status.

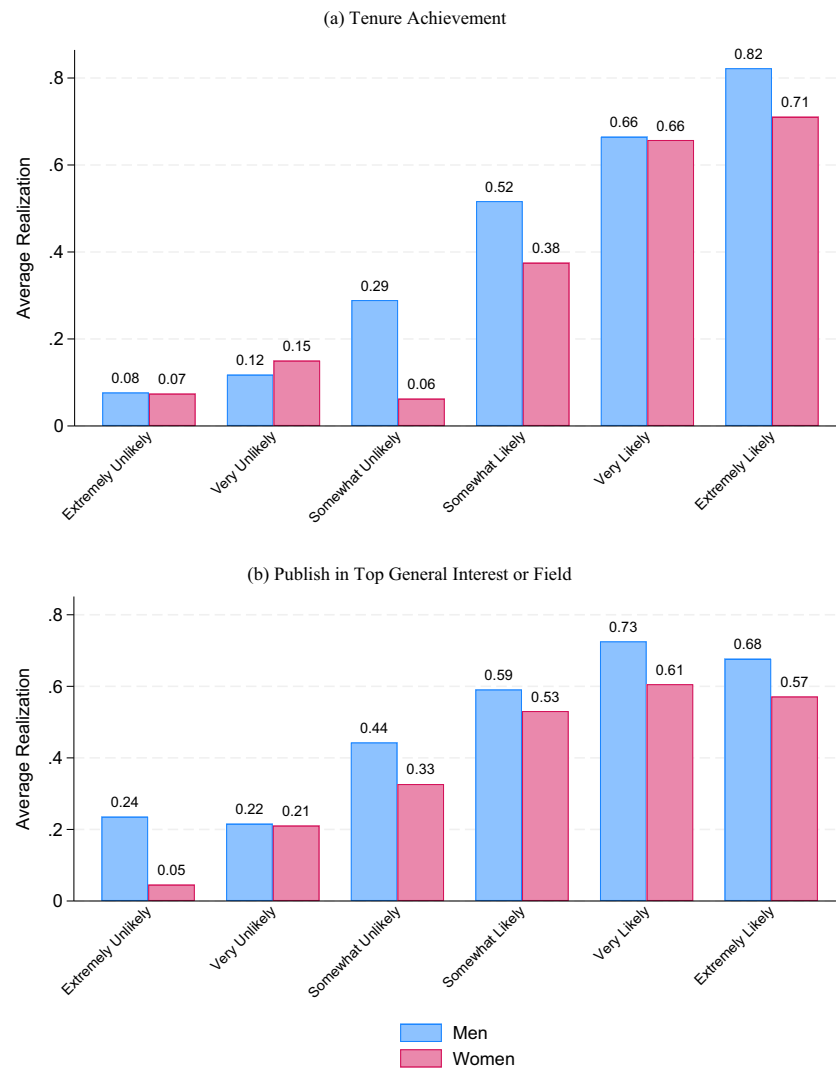


Fig. 1. Average outcome realization by expectation response by gender. Bars represent the mean achievement rate of the given outcome. The publication outcome displayed is an indicator for publishing in either a top general interest journal or top field journal at any point after certificate of PhD.

Table 4
Gender gaps in outcomes.

	(1) Baseline	(2) With relationship status	(3) With controls	(4) With expectations	(5) Controls + expectations
Panel A: receive tenure					
Female	−0.119*** (0.038)	−0.116*** (0.039)	−0.116*** (0.036)	−0.072** (0.035)	−0.091** (0.036)
In a committed relationship		0.058 (0.043)	0.052 (0.042)	0.058 (0.040)	0.054 (0.040)
Expectation				0.404*** (0.033)	0.259*** (0.041)
Panel B: publish in top journals					
Female	−0.104*** (0.033)	−0.099*** (0.034)	−0.101*** (0.033)	−0.080** (0.034)	−0.087*** (0.033)
In a committed relationship		0.068* (0.037)	0.055 (0.035)	0.071* (0.036)	0.059* (0.035)
Expectation				0.164*** (0.040)	0.119*** (0.038)
Observations	769	769	769	769	769

Sample is restricted to participants with expectation and outcome data for both tenure achievement and publishing. Job preferences and field of study are measured in the pre-market survey. Accepted positions are measured in the post-market survey (which is the expectation survey for the third Cohort) and supplemented by our own data collection when missing. Age, relationship status, citizenship, and race are measured at the time of the expectation survey. The indicator for PhD from a top program is comprised of 39 U.S. and international programs, detailed in Appendix B. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the gender gap. For publishing, expectations account for 1.4 percentage points of the gap, or about 14 % of the gender gap.⁸

Appendix Table A5 presents the analogous table with career citations as the outcome variable. The only difference in this specification is that both expectations are included, as publishing and tenure expectations may feasibly be reflected in citation counts. Comparing columns (3) and (5), we see that the inclusion of career expectations results in a reduction of 35.6 % in the gender gap.

3.4. Individually preferred outcomes

Finally, our data offer insights into whether the participant's job market outcome was constrained by their relationship. The post-market survey, which was fielded directly after the job market for all cohorts, asked participants the following:

Please think about the set of jobs from which you chose, including all of your formal offers and any offers you are certain you could have received, but did not formally receive. If your [significant other] would have been equally happy with any outcome of your job search, would you have accepted your [accepted position], or would you have accepted a different job?

If a respondent indicates that they would have accepted the same position, we classify them as having accepted their individually preferred (IP) job choice. For the first two cohorts, this question is asked in a different survey than the one in which expectations are elicited, and thus restricts the sample further than the previous analysis (namely to those with data on expectations, outcomes, and whether their accepted position was individually preferred). These restrictions produce a smaller sample size. While these reduced sample sizes are too small to draw convincing inferences, given the novelty of this question, we still describe the responses.

In total, 402 respondents in our analysis sample responded to this question, of whom 29 % are female. 51 (12.7 %) respondents indicated they rejected their individually preferred (IP) job choice due to their relationship. Of these, 49 took a different job while 2 rejected all jobs. By gender, 33 of 285 (11.6 %) men rejected their IP position and 18 of 117 women (15.4 %) rejected their IP position (p -value for the equality of these proportions by gender is 0.299).

It is worth noting that we find no evidence of tenure expectations and outcomes differing across men according to whether or not they accepted their IP outcome (see Table A6). For women, however, we find substantial gaps by IP option choice: women who rejected their IP option are 14 percentage points less likely to believe tenure achievement is likely and 16 percentage points less likely to end up achieving tenure. Neither of these differences is statistically significant, however (and is based on very small sample sizes).

This analysis, while underpowered, suggests that intra-household constraints may appear to be more binding for women, and more consequential for their subsequent long-term professional outcomes.

3.5. Robustness

3.5.1. Timing of expectation elicitation

As mentioned in Section 2.1, expectations were elicited 3–4 years after the job market (for Cohorts 1 and 2), or immediately following the job market (Cohort 3). What we really need are expectations at the onset of one's career. One concern is that expectations for the first two cohorts may have already evolved over the course of the first few years differentially by gender. As a robustness check, we use only the third cohort, for whom the timing of expectations data elicitation is the closest to the start

of the professional career. Appendix Table C1 presents ex-ante expectations and ex-post outcomes, analogous to Table 2, restricting the sample to Cohort 3.⁹ While the sample size is necessarily reduced resulting in lower precision, we see that the differences by gender are qualitatively similar. Women's expectations for getting tenure and publishing regularly are, on average, lower by 17.1 and 14.2 percentage points than males' expectations, respectively. Gender gaps in outcomes are smaller in magnitude but economically sizable.

It is also worth noting that, controlling for expectations, the gender gap in tenure and publications is reduced by 17 % and 10 % respectively for this cohort. These are remarkably similar to the gender gap explained by expectations in the full sample (in Table 4).

3.5.2. Academic placement

The expectations and outcomes we analyze are inherently academic in nature. One would expect that somebody who initially places into a private sector position is more likely to have a preference for the private sector, and is unlikely to pursue publication or tenure. This should not be an issue since expectations are elicited after the initial placement is known, and thus participants should be taking their placement into account. In addition, our analysis includes whether initial placement is in academia as one of the controls.

However, as a robustness check, we re-run the analysis further restricting the sample to those who had an initial placement that was academic.¹⁰ Appendix Table C2 presents the analogue to Table 3, restricted to this sample which had an initial academic placement. Our sample is necessarily reduced, decreasing precision as not all participants' initial placements could be confidently identified. Qualitatively, the results are unchanged with women 10.1 and 12.7 percentage points less likely to expect to receive tenure and publish regularly, respectively. Women are also 11.0 and 12.7 percentage points less likely to receive tenure and publish, respectively, with the differences significant and robust to controls.

3.5.3. Relationship timing

Throughout the main analysis, we use relationship status as defined at the time expectations were elicited. It is reasonable to expect that individuals internalize their current relationship status at the time of forming their ex-ante career expectations. It is possible, however, that decisions made in the past due to relationship constraints play a more significant role in expectation formation. We investigate this by re-running the entirety of the analysis using an alternative definition of relationship status, measured at the time of the pre-market survey which occurred at the beginning of the job market. The presence of a relationship at this time places potential constraints on location choices, interviews and fly outs, and the eventual accepted position.

Performing this analysis reduces our sample size slightly, from 769 participants to 713 (93 %), as we have to impute values for those who did not complete the pre-market survey but did complete the post-market or expectations survey. It is important to note that a majority of those lost in using this alternative definition come from the third cohort. This stems from changes made to how the survey asks participants about prior relationships.

⁸ If instead we control for the entire Likert-scale fixed effects instead of the binary measure of expectations (that is indicators for each of the answer options), the proportion of the gender gap that is explained by expectations is 2.9 percentage points (25 % of the gender gap) for tenure, and 0.6 percentage points (6 %) for publishing.

⁹ It is notable that the level of expectations for this cohort is quite similar to those for the full sample in Table 3, despite the fact that the first cohort entered the market right at the onset of the Great Recession. This suggests that expectations were not drastically affected by the aggregate shock.

¹⁰ The post-market survey explicitly asked about initial placement, but there are participants in the first and second cohort who completed the early-career survey in which expectations are elicited, but did not complete the post-market survey. We complement the original survey data with our own data collection on job history. Participants were coded as having an initial academic placement if they placed into a post-doc or any position within a university – both tenure and non-tenure track.

Appendix D includes additional information on how relationship status at the time of job market is defined as well as the associated analysis. The results using this definition, presented in Appendix D, are qualitatively unchanged. Both men and women are more likely to be single at the time of the job market, with 25 % of men and 34 % of women single at the pre-market survey compared to 21 % of men and 27 % of women at the time of the expectation survey. Table D2 presents the levels of expectations and outcomes by gender and relationship status. Looking at the specifications for differences with controls, which include relationship status, women are 10.8 percentage points and 12.1 percentage points less likely to expect to get tenure and to publish regularly, respectively. The gaps persist in outcomes, with women 11.9 percentage points less likely to receive tenure and 10.1 percentage points less likely to publish regularly. Consistent with the main analysis, we also see weak evidence that not-single men and women do better in outcomes compared to their single counterparts, though these differences are not statistically significant. Expectations do not differ across relationship status within gender, with the differences also not being statistically significant.

4. Conclusion

This paper links novel data on early-career expectations from PhD recipients in economics and adjacent fields with professional outcomes more than a decade later. This unique data then allows us to document several novel facts. Not only do we see that early-career expectations differ systematically by gender in this sample of high-ability high-ambition individuals, but they are also predictive of ex-post outcomes. In addition, we show that expectations can explain about 22 %, 14 %, and 36 % of the gender gap in tenure, publication, and citation outcomes, respectively.

While there is a large literature showing that expectations are predictive of human capital choices and outcomes, almost all of this literature looks at outcomes over shorter horizons. In addition, there is even less work on gender gaps in expectations and outcomes among individuals with professional degrees beyond a bachelor's.

Our analysis raises new questions. It is not clear why expectations differ by gender. It could be because females anticipate discrimination or that they have internalized other social norms (Altonji & Blank, 1999; Wiswall & Zafar, 2021), or this could be due to other factors. Understanding the drivers of gender differences in expectations is needed to inform policies aimed at mitigating gender gaps in outcomes. In addition, we have expectations data at only one point in time. How these career expectations evolve over time, and whether they do so differently by gender, is not well understood.

While there is a large literature on how labor market outcomes are associated with or affected by relationship status, we are not aware of prior work that has data on both ex-post labor market outcomes and ex-ante expectations. While our results are largely descriptive, we believe they are informative. Moreover, our result that expectations and outcomes do not differ systematically by relationship status, within gender, is intriguing and merits further investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpubeco.2025.105437>.

Data availability

The authors do not have permission to share data.

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